

Energy-Flow Cosmology: Recovery Conditions and the Λ CDM Limit

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Abstract. Energy-Flow Cosmology (EFC) introduces a regime-dependent modification to the perturbation sector of standard cosmology, parametrised by an entropy-gradient coupling α_{L2} and a transition function $\Theta(z)$. We establish three analytically independent sufficient conditions under which EFC reduces exactly to Λ CDM or General Relativity (GR): (i) the parameter limit $\alpha_{L2} \rightarrow 0$; (ii) the low-entropy regime $S < S_c$; (iii) density saturation $\rho \gg \rho_{\text{crit}}$. These conditions are shown to be mutually consistent and to yield the same limiting behaviour $\mu(k, z) \rightarrow 1$, $\Sigma(k, z) \rightarrow 1$ without residual corrections. EFC therefore contains Λ CDM as a limiting case in the perturbation sector. A full recovery of the background expansion $H(z)$ has not yet been derived and is stated explicitly as an open problem.

Introduction

A necessary condition for any viable extension of Λ CDM is the ability to recover standard cosmology in an appropriate limit. Without this, the theory is not a *generalisation*—it is merely a competing hypothesis.

EFC modifies the growth sector through an effective gravitational coupling that depends on local entropy structure and cosmic epoch. The central question addressed here is whether Λ CDM emerges as a well-defined limit.

We identify three independent limits, show their mutual consistency, and state precisely what remains to be established. This section constitutes the formal anchor of the EFC whitepaper series.

The EFC Perturbation Framework

Core parameterisation

The EFC modification enters the background expansion as:

$$E^2(z) \longrightarrow E^2(z) [1 + \alpha_{L2} f(z) \Theta(z)], \quad (1)$$

where $E(z) \equiv H(z)/H_0$, $f(z)$ encodes the growth-sector coupling, and $\Theta(z)$ is a smooth regime-transition function:

$$\Theta(z) = \frac{1}{2} \left[1 + \tanh \left(\frac{z_{L1L2} - z}{\Delta z} \right) \right]. \quad (2)$$

$\Theta(z) \rightarrow 1$ in the late-time, structure-dominated regime (L2) and $\Theta(z) \rightarrow 0$ at high redshift (L1), ensuring CMB-safe behaviour.

Modified growth functions

In the perturbation sector, EFC introduces scale- and time-dependent modifications to the two standard gravitational slip parameters:

$$\mu(k, z) \equiv \frac{G_{\text{eff}}(k, z)}{G_N} = 1 - B g(a; n), \quad (3)$$

$$\Sigma(k, z) \equiv \frac{G_{\text{lens}}(k, z)}{G_N} = \mu_{\text{lens}}(k, z), \quad (4)$$

with $g(a; n) = [1 + (a_t/a)^n]^{-1}$ a sigmoid gate function and $B > 0$ a dimensionless amplitude. $\mu < 1$ implies weakened gravitational growth, consistently with the observed S_8 tension.

The EFC kernel takes the explicit form:

$$\mu_{\text{EFC}}(k, z) = (1 - \bar{S}(z)) + \frac{\tilde{s}(k, z)}{\bar{\delta}(k, z)}, \quad (5)$$

where $\bar{S}(z)$ is the mean entropy field and $\tilde{s}/\bar{\delta}$ is the scale-dependent entropy–density ratio.

The effective gravitational coupling at the field-equation level reads:

$$G_{\mu\nu} = f(\nabla S, E_f), \quad (6)$$

reducing to the standard Einstein tensor when entropy gradients vanish (see Section 3).

Recovery Conditions and the Λ CDM Limit

Overview

We prove three sufficient conditions, each independently yielding $\mu = \Sigma = 1$ and standard GR/ Λ CDM behaviour. Their mutual consistency is established in Proposition 1.

Condition I: Parameter limit

Theorem 1 (Parameter recovery). *In the limit $\alpha_{L2} \rightarrow 0$, EFC reduces to Λ CDM in the perturbation sector:*

$$\alpha_{L2} \rightarrow 0 \implies \mu(k, z) \rightarrow 1, \quad \Sigma(k, z) \rightarrow 1. \quad (7)$$

Proof. When $\alpha_{L2} = 0$, equation (1) reduces to the standard Friedmann form $E^2(z)$. The EFC correction term $\alpha_{L2} f(z) \Theta(z)$ vanishes identically for all z and all k , so $G_{\text{eff}} = G_N$ and $\mu = \Sigma = 1$ at every scale. $\square$

Observational constraint: The sealed blind prediction (NUTS cycle, 2026-02-18) yields $\alpha_{L2} = -1.00 \pm 0.46$ at 2.20σ with $\Delta\text{AIC} = -2.91$. Seven leave-one-out folds all pass the signal criterion. The non-zero measurement is therefore an empirical deviation from this limit, not a theoretical requirement.

Condition II: Low-entropy regime

Theorem 2 (Entropy recovery). *In the low-entropy regime $S < S_c \approx 0.1$, all EFC corrections vanish and the model reduces to classical General Relativity:*

$$S < S_c \implies \mu(k, z) \rightarrow 1, \quad \Sigma(k, z) \rightarrow 1. \quad (8)$$

Proof. From equation (5), when $\bar{S}(z) \rightarrow 0$ and $\tilde{s}/\tilde{\delta} \rightarrow 0$, we obtain $\mu_{\text{EFC}} \rightarrow 1$. In this regime the entropy field S carries no dynamical content and equation (6) reduces to the standard Einstein tensor $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}$. This is the L0/L1 regime: particle physics and solar-system scales, where EFC is explicitly designed to be inert. $\square$

Observational link: Solar system (PPN) and stellar-interior tests confirm GR recovery quantitatively (validation KT1: Newtonian limit recovered; EFC correction $< 10^{-5}$).

Condition III: Density saturation

Theorem 3 (Density recovery). *Via the density-saturation screening function $\Theta(\rho)$, EFC corrections are suppressed in high-density environments:*

$$\rho \gg \rho_{\text{crit}} \implies \Theta(\rho) \rightarrow 0 \implies G_{\text{eff}} \rightarrow G_N. \quad (9)$$

Proof. The density-saturation function $\Theta(\rho)$ acts as a chameleon-type screening mechanism. In the limit $\rho \gg \rho_{\text{crit}}$, $\Theta(\rho) \rightarrow 0$ and the EFC correction to G_{eff} is driven to zero. This recovers standard GR in Solar System interiors and dense stellar environments, as required by post-Newtonian constraints. $\square$

Observational link: Grid-AQUAL kill-test KT2 confirms the prefactor convergence $C_{vw} = 2.2960$; KT3 establishes Λ -locking with $\beta \approx 0.18$.

Mutual consistency of the three limits

Proposition 1 (Triangulated recovery). *The three sufficient conditions — parameter limit, entropy limit, and density saturation — yield the same limiting behaviour without mutual conflict:*

$$[\alpha_{L2} \rightarrow 0] \vee [S < S_c] \vee [\rho \gg \rho_{\text{crit}}] \implies \mu(k, z) \rightarrow 1, \quad \Sigma(k, z) \rightarrow 1, \quad (10)$$

for all scales k and redshifts z in the perturbation sector.

Proof. Each condition independently drives the EFC correction term to zero through a distinct physical channel: (I) via the coupling amplitude, (II) via the entropy source field, (III) via local density suppression. There are no cross-terms between the three channels that could generate residual corrections when any one condition holds. The three limits are therefore mutually consistent and non-conflicting. $\square$

Corollary 1 (EFC $\supset$ Λ CDM). *EFC contains Λ CDM as a limiting case in the perturbation sector.*

$$\text{EFC} \supset_{\text{pert.}} \Lambda\text{CDM}$$

Scope Boundaries and Open Problems

What has been established

Table 1 summarises the three recovery channels with their physical mechanism, validated domain, and status.

Table 1: EFC recovery conditions: status summary

Condition	Limit	Domain	Status
(I) Parameter	$\alpha_{L2} \rightarrow 0$	Full perturbation sector	Analytic
(II) Entropy	$S < S_c \approx 0.1$	L0/L1 (GR regime)	Analytic + KT1
(III) Density	$\rho \gg \rho_{\text{crit}}$	Solar system, stellar	Analytic + KT2/KT3

Table 2: EFC $\supset$ Λ CDM: level classification

Level	Claim	EFC status
1	Compatible with Λ CDM in some regimes	Established
2	Λ CDM as a limiting case (perturbation sector)	Established (this work)
3	Full generalisation incl. background $H(z)$	Open problem

Explicit open problem

Remark 1 (Background recovery — partial progress in Part 4). *No consistent mapping from the EFC formulation to the standard background expansion $H(z)$ has been established at the Lagrangian level. Specifically, the background α -gate on the Friedmann equation was found to be inconsistent with joint CMB+BAO constraints. EFC therefore operates as a perturbation-sector modification only at the level of Parts 1–3.*

Partial progress toward this derivation is established in Part 4 of this series via the regime susceptibility function $T(S) = S_0(1 - S_0)/[S(1 - S)]$ and the candidate equation of state $w(a) = -\beta(S) \cdot a$ [10]. A full Level 3 claim additionally requires that Ω_m and Ω_{DE} be derived from the EFC entropy field rather than imported from observations. This remains an open problem.

This boundary is a feature, not a defect: it precisely defines the domain of validity and protects the empirical claims from overreach.

Discussion

Structural analogy

The recovery structure of EFC mirrors the limiting behaviour of successful physical generalisations: GR reduces to Newtonian gravity in the limit $v/c \rightarrow 0$; QFT reduces to classical field theory in the limit $\hbar \rightarrow 0$. EFC recovers Λ CDM in the limit $\alpha_{L2} \rightarrow 0$ and $S \rightarrow 0$. This is the standard criterion for a framework that *extends* rather than *replaces* its predecessor.

Implications for the degeneracy problem

Parameter α_{L2} is not uniquely identifiable from fewer than three independent probes (identifiability gate IG-1, collapsed under two-probe combinations). This is not a defect of the theory: it is a structural consequence of the μ – Σ degeneracy sector. The remedy is multi-probe consistency (DES + KiDS + DESI + BOSS simultaneously), which is precisely what the EFC validation pipeline is designed to achieve.

Falsification criterion

The recovery theorems generate a sharp falsification criterion:

EFC is falsified if $\mu(k, z) = \Sigma(k, z) = 1$ to within measurement precision across all probes simultaneously, while the individual data constraints are tight enough to exclude $|\alpha_{L2}| > 0.1$ at 3σ .

This is a stricter criterion than the frozen Euclid/Rubin LSST kill-test (DOI: 10.6084/m9.figshare.31251277).

Conclusions

We have established three analytically independent sufficient conditions under which Energy-Flow Cosmology reduces to Λ CDM or General Relativity. The three conditions are mutually consistent (Proposition 1), yielding the containment relation $\text{EFC} \supset_{\text{pert.}} \Lambda\text{CDM}$ (Corollary 1). This result upgrades EFC from a *competing alternative* to a *generalising framework* with correct limiting behaviour in the perturbation sector. The open problem — full background $H(z)$ recovery — is stated precisely and delineates the boundary of the current claim.

Reproducibility.

All sealed blind predictions referenced here carry cryptographic hashes: freeze 20260218 (hash 7a850cfa...) and freeze 20260221 (hash dbccda15...).

The full validation ledger is publicly available at https://supertedai.github.io/EFC/public/EFC_Validation_Ledger.html.

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